

Formatting thread: `::id` and `stacktrace`

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Abstract

This paper provides wording in reply to NB comments suggesting to adopt [P1636R2](#) [1] (Formatters for library types) and to add formatters for `std::stacktrace`.

Revisions

R1

- Wording fixes following LWG feedback
- Question to LEWG: should `stacktrace_entry` formatting support align and fill options? LWG argue that it may be useful. Because it requires adding state to the formatter it would be an ABI break if added later. Adding it now cost very little so if it may be useful to someone, we might as well (even if we don't have a compelling use case to illustrate).

History

LEWG approved [P1636R2](#) [1] in 2019 (Cologne) for C++20. The paper was subsequently reviewed in 2021 by LWG, who requested wording changes.

SG16 had significant concerns with the formatting of `filesystem::path` and asked for that formatter to be removed.

Another paper [P2197R0](#) [2] explored different options for formatting `std::complex` but was not pursued, nor was it warmly received by LEWG when presented (summer 2020 telecon).

[P1636R2](#) [1] has been stuck in need of a revision and attempts to contact the author have failed.

Design

This paper provides wording for

- `std::thread::id`
- `std::basic_stacktrace`
- `std::stacktrace_entry`

Note that [P1636R2 \[1\]](#) additionally proposed to support

- `std::complex`
- `std::bitset`
- `std::error_code`
- `std::unique_ptr`
- `std::shared_ptr`

Which we decided not to pursue as part of this NB comment resolution.

Why do we need `thread::id` formatting in C++23?

Two reasons. First, it is very commonly used by loggers. But most importantly that information is not exposed by any other means than an `ostream` << overload. There is no accessor of any kind, so the only well-defined way to extract a `thread::id` is to use streams

```
std::ostringstream ss;
ss << thread.get_id();
std::print("called a nice API on thread {}", ss.str());
```

Note that a quick search on GitHub reveals that users, when their expectations are subverted, will hm... find a way, and won't let such thing as well-definedness stop them. They will, for example, exploit the amazing flexibility of `printf` to get what they want:

```
printf("[Thread %d Profiling: %ld microseconds] ",
      std::this_thread::get_id(), microseconds); // UB
```

By properly supporting `thread::id` in format, we can avoid the proliferation of undefined, non portable, and dangerous code.

Why not `std::complex`?

Formatting of `std::complex` is more... complex. In particular, [P1636R2 \[1\]](#) (which, to be fair, was approved by LEWG) proposed to use the same notation as `ostream`. Given the wide variance in how `complex` are formatted in other programming languages and the interaction with locales (including the need to support the `L` specifier), it seems wise to punt this question to a later C++ version.

ostream	(1.0, 2.0)
P1636R2 [1]	(1.0, 2.0)
P2197R0 [2]	(1.0, 2.0i)
Rust	1+2i
Python	1+2j
Julia	1.0+ 2.0im
Nim	(1.0, 2.0)
R	1+2i
Swift	(1.0, 2.0)
C#	1 + 2i

Why not `std::error_code/bitset/smart pointers`?

These don't seem sufficiently useful to be processed as part of NB comments.

Moreover, there were plans to remove smart pointer formatters from P1636 for consistency with raw pointers which are intentionally not formattable by default.

Stacktrace

We propose adding formatters for `std::stacktrace` and `std::stacktrace_entry` in addition to existing `std::to_string` overloads such that the following would be equivalent:

```
to_string(a_stacktrace);
std::format("{} ", a_stacktrace);
```

Do we really need both? Beside arithmetic types, the only other type to have a `std::to_string` method is `std::bitset`.

Should stacktrace be formatted as a range?

This would add a lot of complexity to something that would probably never be used. The range behavior can be opt-in this way instead:

```
std::format("{} ", std::views::all(a_stacktrace));
```

Wording

Header `<stacktrace>` synopsis

[stacktrace.syn]

```
#include <compare>           // see ??

namespace std {
    // ??, class stacktrace_entry
    class stacktrace_entry;
```

```

// ??, class template basic_stacktrace
template<class Allocator>
class basic_stacktrace;

// basic_stacktrace typedef-names
using stacktrace = basic_stacktrace<allocator<stacktrace_entry>>;

// ??, non-member functions
template<class Allocator>
void swap(basic_stacktrace<Allocator>& a, basic_stacktrace<Allocator>& b)
noexcept(noexcept(a.swap(b)));

string to_string(const stacktrace_entry& f);

template<class Allocator>
string to_string(const basic_stacktrace<Allocator>& st);

template<class charT, class traits>
basic_ostream<charT, traits>&
operator<<(basic_ostream<charT, traits>& os, const stacktrace_entry& f);

template<class charT, class traits, class Allocator>
basic_ostream<charT, traits>&
operator<<(basic_ostream<charT, traits>& os, const basic_stacktrace<Allocator>& st);

// ??, formatting support
template<> struct formatter<stacktrace_entry>;
template<class Allocator> struct formatter<basic_stacktrace<Allocator>>;

namespace pmr {
    using stacktrace = basic_stacktrace<polymorphic_allocator<stacktrace_entry>>;
}

// ??, hash support
template<class T> struct hash;
template<> struct hash<stacktrace_entry>;
template<class Allocator> struct hash<basic_stacktrace<Allocator>>;
}

template<class charT, class traits, class Allocator>
basic_ostream<charT, traits>&
operator<<(basic_ostream<charT, traits>& os, const basic_stacktrace<Allocator>& st);

```

Effects: Equivalent to: return os << to_string(st);

[...]

Formatting support

[stacktrace.format]

```
template<> struct formatter<stacktrace_entry>;
```

formatter<stacktrace_entry> interprets *format-spec* as a *stacktrace-entry-format-spec*. The syntax of format specifications is as follows:

stacktrace-entry-format-spec:
*fill-and-align*_{opt} *width*_{opt}

[*Note*: The productions *fill-and-align* and *width* are described in [format.string]. — *end note*]

A stacktrace_entry object *se* is formatted as if by copying to_string(*se*) through the output iterator of the context with additional padding and adjustments as specified by the format specifiers.

```
template<class Allocator> struct formatter<basic_stacktrace<Allocator>>;
```

For formatter<basic_stacktrace<Allocator>>, *format-spec* is empty.

A basic_stacktrace object *s* is formatted as if by copying to_string(*s*) through the output iterator of the context.

◆ Hash support [stacktrace.basic.hash]

```
template<> struct hash<stacktrace_entry>;  
template<class Allocator> struct hash<basic_stacktrace<Allocator>>;
```

The specializations are enabled[unord.hash].

◆ Concurrency support library [thread]

◆ Class thread::id [thread.thread.id]

```
namespace std {  
    class thread::id {  
    public:  
        id() noexcept;  
    };  
  
    bool operator==(thread::id x, thread::id y) noexcept;  
    strong_ordering operator<=>(thread::id x, thread::id y) noexcept;  
  
    template<class charT, class traits>  
    basic_ostream<charT, traits>&  
    operator<<(basic_ostream<charT, traits>& out, thread::id id);  
  
    // hash support  
    template<class T> struct hash;  
    template<> struct hash<thread::id>;  
  
    template<class charT>  
    struct formatter<thread::id, charT>;  
}
```

An object of type `thread::id` provides a unique identifier for each thread of execution and a single distinct value for all thread objects that do not represent a thread of execution[`thread.thread.class`]. Each thread of execution has an associated `thread::id` object that is not equal to the `thread::id` object of any other thread of execution and that is not equal to the `thread::id` object of any thread object that does not represent threads of execution.

The *text representation* for the character type `charT` of an object of type `thread::id` is an unspecified sequence of `charT` such that, for two objects of type `thread::id` `x` and `y`, if `x == y` the `thread::id` objects have the same text representation and if `x != y` the `thread::id` objects have distinct text representations.

`thread::id` is a trivially copyable class[`class.prop`]. The library may reuse the value of a `thread::id` of a terminated thread that can no longer be joined.

Let $P(x,y)$ be an unspecified total ordering over `thread::id` as described in **??**.

Returns: `strong_ordering::less` if $P(x,y)$ is true. Otherwise, `strong_ordering::greater` if $P(y,x)$ is true. Otherwise, `strong_ordering::equal`.

```
template<class charT, class traits>
basic_ostream<charT, traits>&
operator<< (basic_ostream<charT, traits>& out, thread::id id);
```

Effects: Inserts ~~an unspecified~~ the text representation for `charT` of `id` into `out`. ~~For two objects of type `thread::id` `x` and `y`, if `x == y` the `thread::id` objects have the same text representation and if `x != y` the `thread::id` objects have distinct text representations.~~

Returns: `out`.

```
template<class charT> class formatter<thread::id, charT>;
```

`formatter<thread::id, charT>` interprets *format-spec* as a *thread-id-format-spec*. The syntax of format specifications is as follows:

thread-id-format-spec:
 $\text{fill-and-align}_{opt} \text{width}_{opt}$

[*Note:* The productions *fill-and-align* and *width* are described in [format.string]. — *end note*]

If the *align* option is omitted it defaults to `>`.

A `thread::id` object is formatted by writing its text representation for `charT` to the output with additional padding and adjustments as specified by the format specifiers.

Feature test macro

[Editor's note: define `__cpp_lib_formatters` set to the date of adoption in `<version>`, `<stacktrace>` and `<thread>`].

Acknowledgments

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References

- [1] Lars Gullik Bjønnes. P1636R2: Formatters for library types. <https://wg21.link/p1636r2>, 10 2019.
- [2] Michael Tesch and Victor Zverovich. P2197R0: Formatting for std::complex. <https://wg21.link/p2197r0>, 8 2020.
- [N4885] Thomas Köppe *Working Draft, Standard for Programming Language C++* <https://wg21.link/N4885>